

Test Driven Development

“All code is guilty until proven innocent”

Toby Weston
Ben
Lithgow-Smith
Pete Suggitt



Learning Objectives

- Understanding the basic **mechanics** of TDD
- Understanding the power of **Pair Programming**
- Gain insight into the power of the IDE (forward generating code)
- Understand TDD as a **design** tool



WHAT'S YOUR
STORY

"Our highest priority is to satisfy the customer through early and continuous delivery of valuable software"

Principle 1, The Agile Manifesto



“ **Inspection does not improve the quality, nor guarantee quality.** Inspection is **too late**. The quality, good or bad, is already in the product. As Harold F. Dodge said, ‘**You cannot inspect quality into a product.**’ ”

W. Edwards

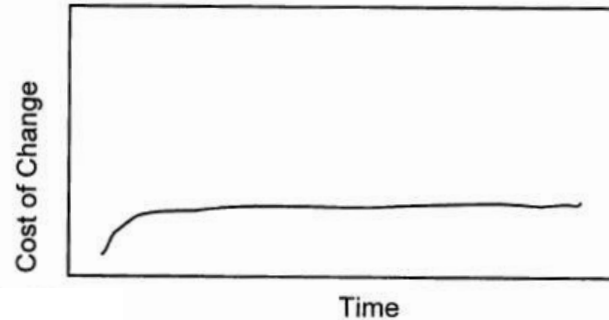
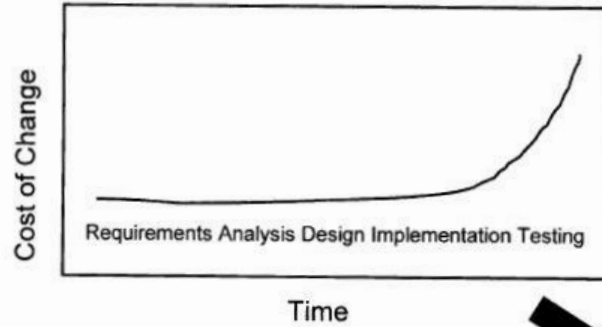
Deming

Out of the Crisis





Cost of Change



Test Driven Development

Write a
failing
test

Write a
failing
test



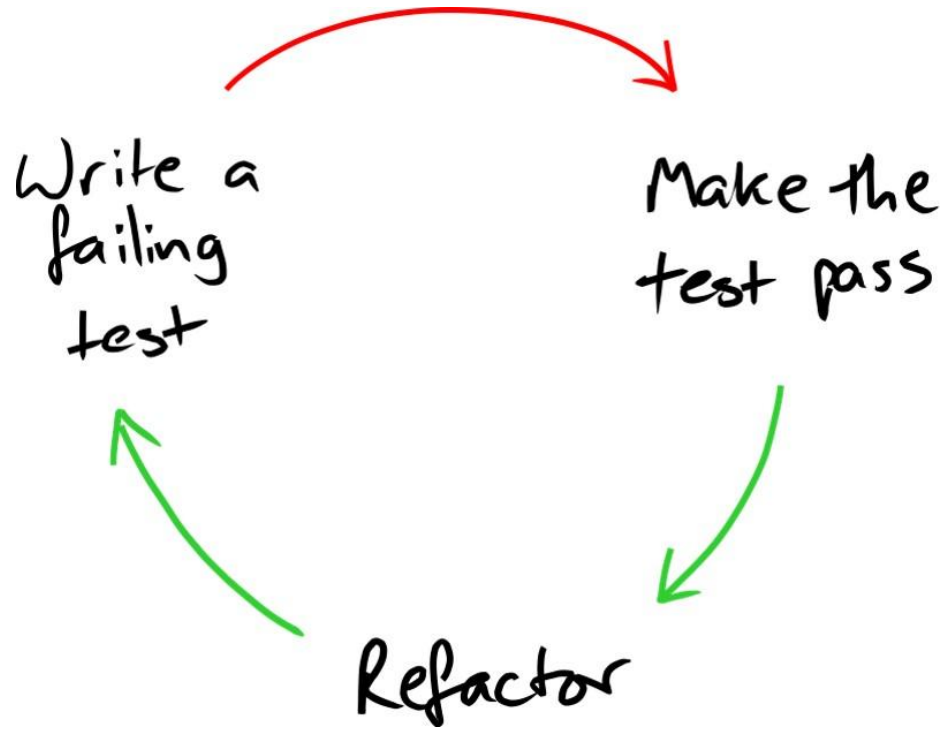
Make the
test pass

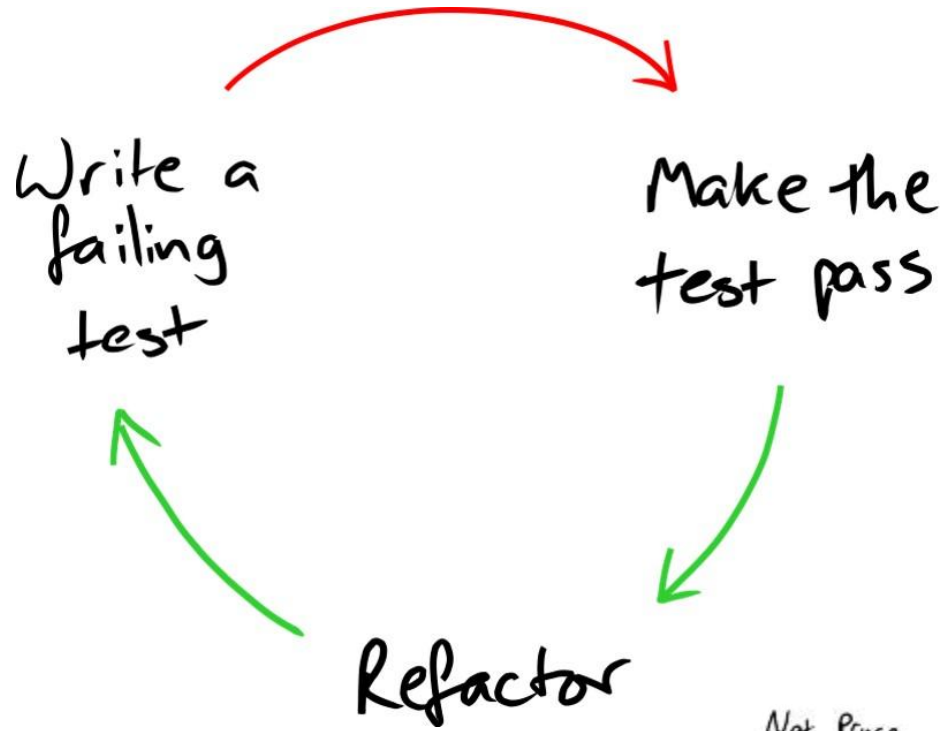
Write a
failing
test

Make the
test pass

Refactor







Nat Pryce

Laws of TDD

You must write a failing test before you write any production code

You must not write more of a test than is sufficient to fail, or fail to compile

You must not write more production code than is sufficient to make the currently failing test pass



“ The act of writing a unit test is more an act of **design** than of verification. It is also more an act of **documentation** than of verification. The act of writing a unit test closes a remarkable number of **feedback loops**, the **least of which is** the one pertaining to **verification** of function. ”

Robert C. Martin

Uncle Bob





Rules of Thumb

Tests capture ***intent***, and test ***one*** thing

Focus on ***what*** is needed; not ***how*** to achieve it

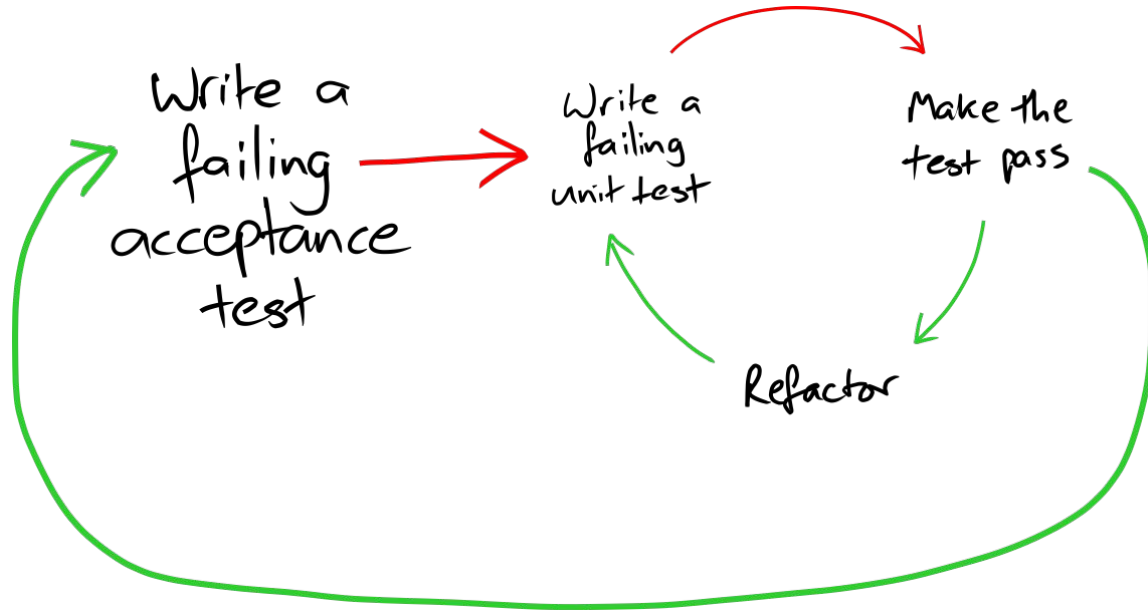
Write the code you want to read

Think KISS / YAGNI / DRY

Arrange, Act, Assert / Given, When, Then



Acceptance Test Driven Development





```

21712 function(scope, element, attr, ngSwitchController) {
21713   // scope
21714   selectedTranscludes = [],
21715   selectedElements = [],
21716   previousElements = [],
21717   selectedScopes = [];
21718
21719   scope.$watch(watchExpr, function ngSwitchWatchAction(value) {
21720     // i, ii
21721     for (i = 0, ii = previousElements.length; i < ii; ++i) {

```

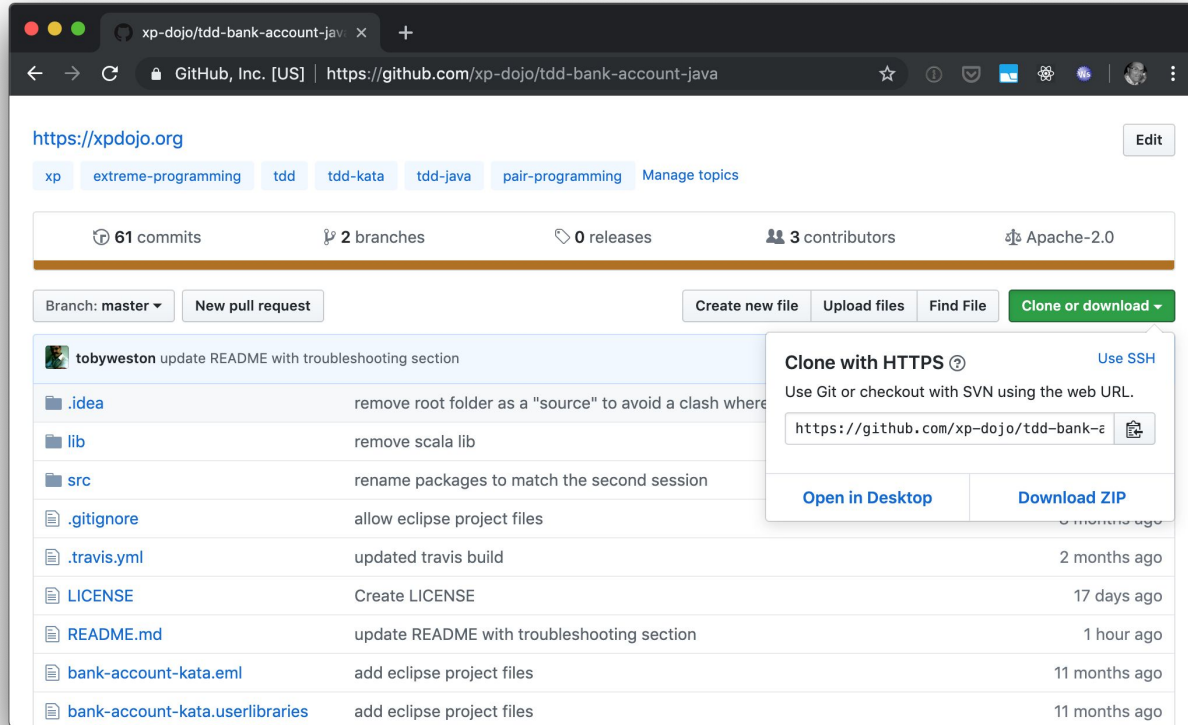
Let's Code!

```

21722     selectedScopes[i].$destroy();
21723     previousElements[i] = selected;
21724     $animate.leave(selected, function() {
21725       previousElements.splice(i, 1);
21726     });
21727   });
21728
21729   selectedElements.length = 0;
21730   selectedScopes.length = 0;
21731
21732   if ((selectedTranscludes = ngSwitchController.cases['!' + value] || ngSwitchController.defaultCase)) {
21733     scope.$eval(attr.change);
21734     forEach(selectedTranscludes, function(transclude) {

```


https://github.com/xp-dojo/tdd-bank-account-java



IntelliJ: “File->Open” and select the folder

Eclipse: “File->Open Projects from File System...” and select directory

User Requirements

- I can **Deposit** money into an account
- I can **Withdraw** money from an account
- I can **Transfer** money between accounts (if I have the funds)
- I can print out the last account **Transaction** (date, amount, balance)
- I can print a **Statement** of account activity (all transactions)
- I can apply statement **Filters** (just deposits, withdrawals, date)





